

Scientific Poster Design

Turning your cancer genomics analysis into a clear research story

Breast Cancer Genomics Project



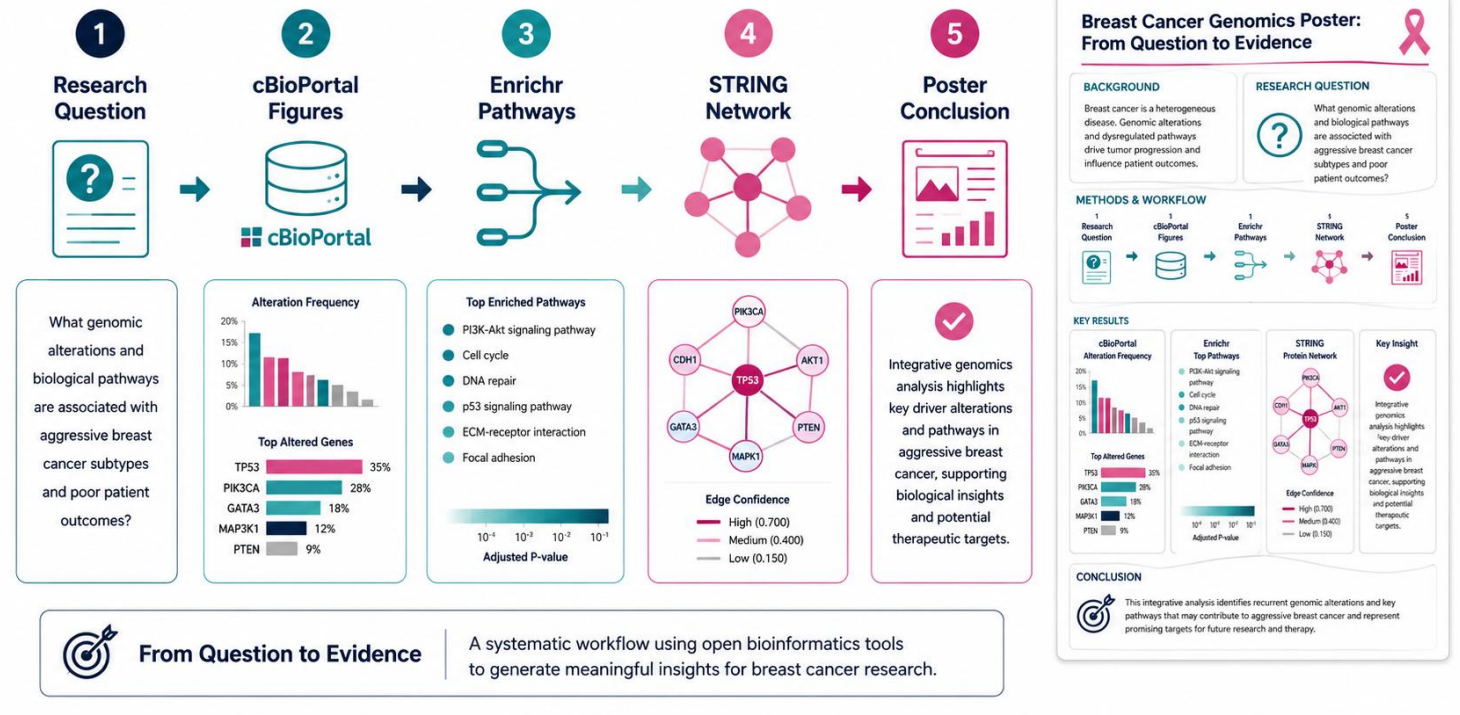
Today's goal

Move from “I made some figures” to “I can explain a scientific result.”

By the end, you should be able to

- Choose figures that support your question
- Write short results statements
- Organize a poster with a clear flow
- Explain limitations honestly

Breast Cancer Genomics Poster: From Question to Evidence



A poster is an argument, not a scrapbook

CORE IDEA

Every figure should answer part of the research question.

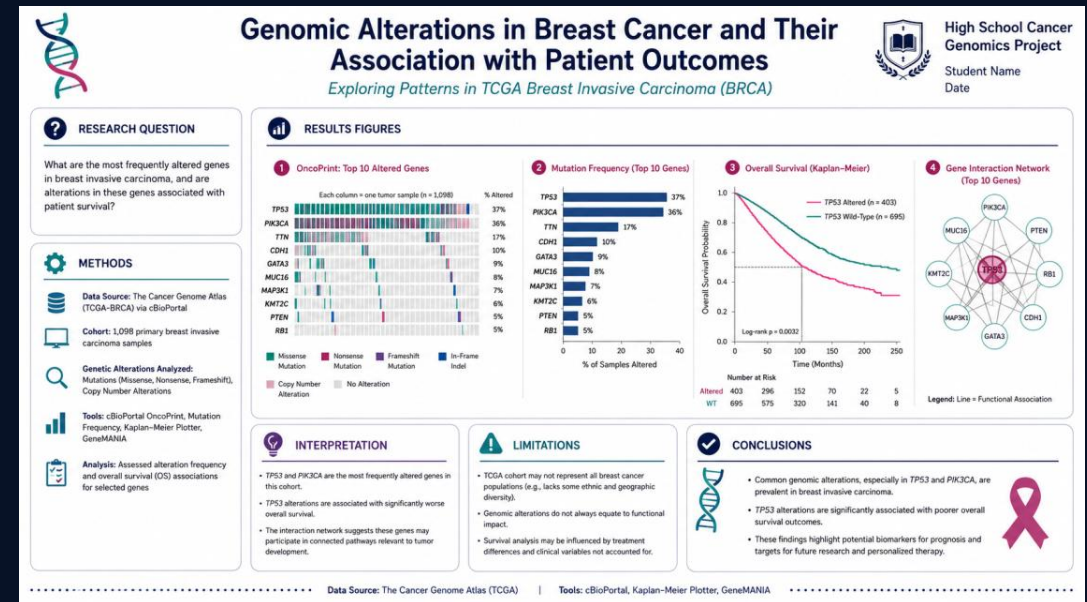
A strong poster says: “Here is my question, here is my evidence, and here is what the evidence means.”

Not the goal

- Include everything you did
- Show every screenshot
- Use tiny text to fit more

The real goal

- Select the best evidence
- Explain the pattern
- Make the reader remember one result



The poster story arc

STRUCTURE

Use the same logic as a short research paper.

1

Question

What comparison are you making?

2

Approach

What data and tools did you use?

3

Evidence

What figures answer the question?

4

Interpretation

What does the pattern suggest?

5

Limits

What can the data not prove?

Required poster sections

Use the same section names across groups so posters are easy to evaluate.

Opening sections

- Title: specific and short
- Background: why the question matters
- Research question: one clear sentence

Methods

- Dataset: TCGA PanCancer Atlas BRCA
- Groups and genes analyzed
- Tools: cBioPortal, Enrichr, STRING

Results

- cBioPortal figure
- Pathway result
- STRING network
- One table or summary chart

Conclusion/interpretation

- What biological pattern do the figures support?
- What does it suggest about subtype biology or target priority?

Limitations and next step

- What can this analysis not prove?
- What experiment or analysis would you do next?

Start with the question

QUESTION → TITLE

Your title should come from your research question, not from the tool you used.

Weak title

- Breast Cancer Project
- cBioPortal Analysis
- Looking at Cancer Genes

Stronger title

- TP53 Alteration Patterns Distinguish Basal-like and Luminal A Breast Cancer
- PIK3CA Alterations Support Luminal A Subtype-Specific Target Prioritization
- ERBB2 Amplification Defines a Targetable HER2-Enriched Subgroup

Methods should be reproducible, not long

METHODS

A reader should be able to repeat your analysis from your Methods box.

Include

- Dataset name
- Comparison groups
- Gene list
- Tools used
- Date accessed

Do not overdo

- Every click you made
- Long tool history
- Unnecessary definitions
- Raw data tables

Example: “We compared BRCA_Basal and BRCA_LumA samples in the TCGA PanCancer Atlas breast cancer study using cBioPortal Group Comparison.”

Methods rule

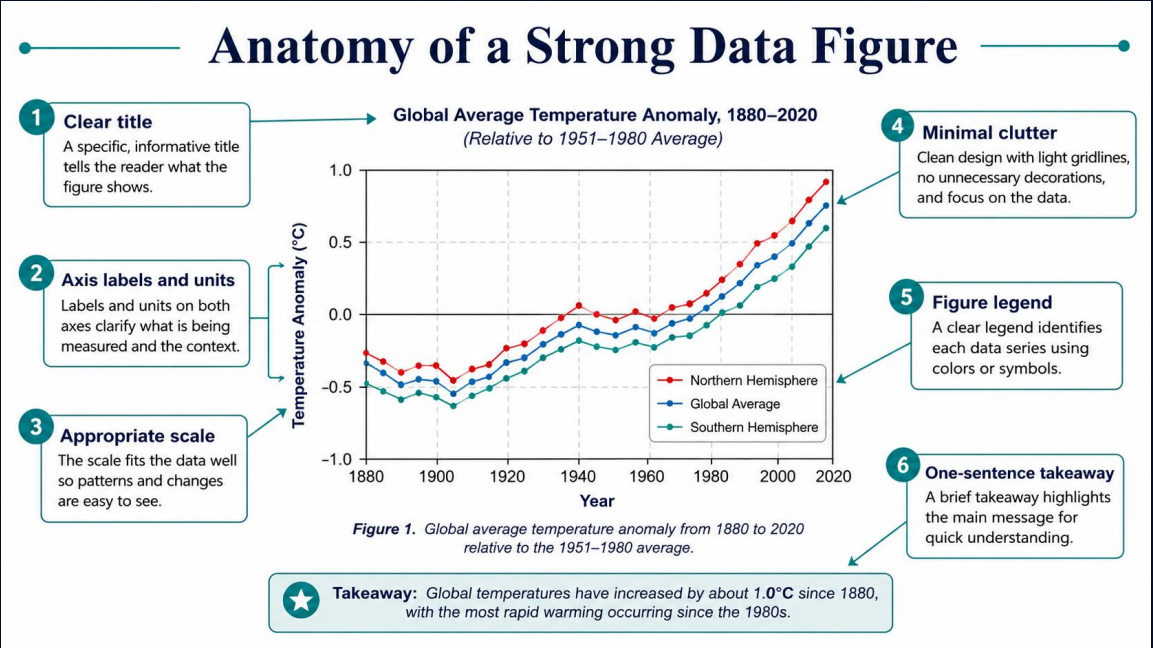
- Enough detail to reproduce the analysis
- Not so much detail that the result disappears

Choose figures by purpose

FIGURE CHOICE

Do not include a figure unless you can state exactly what it proves or suggests.

Question type	Best figure
Cohort description	Subtype chart
Subtype difference	Genomic Alterations comparison
Gene panel pattern	OncoPrint
Mutation details	Mutations Beta lollipop plot
Outcome association	Kaplan–Meier survival curve
Pathway meaning	Enrichr bar/table
Protein relationships	STRING network



Every figure needs a message

A figure without a takeaway is decoration.

FIGURE ANATOMY

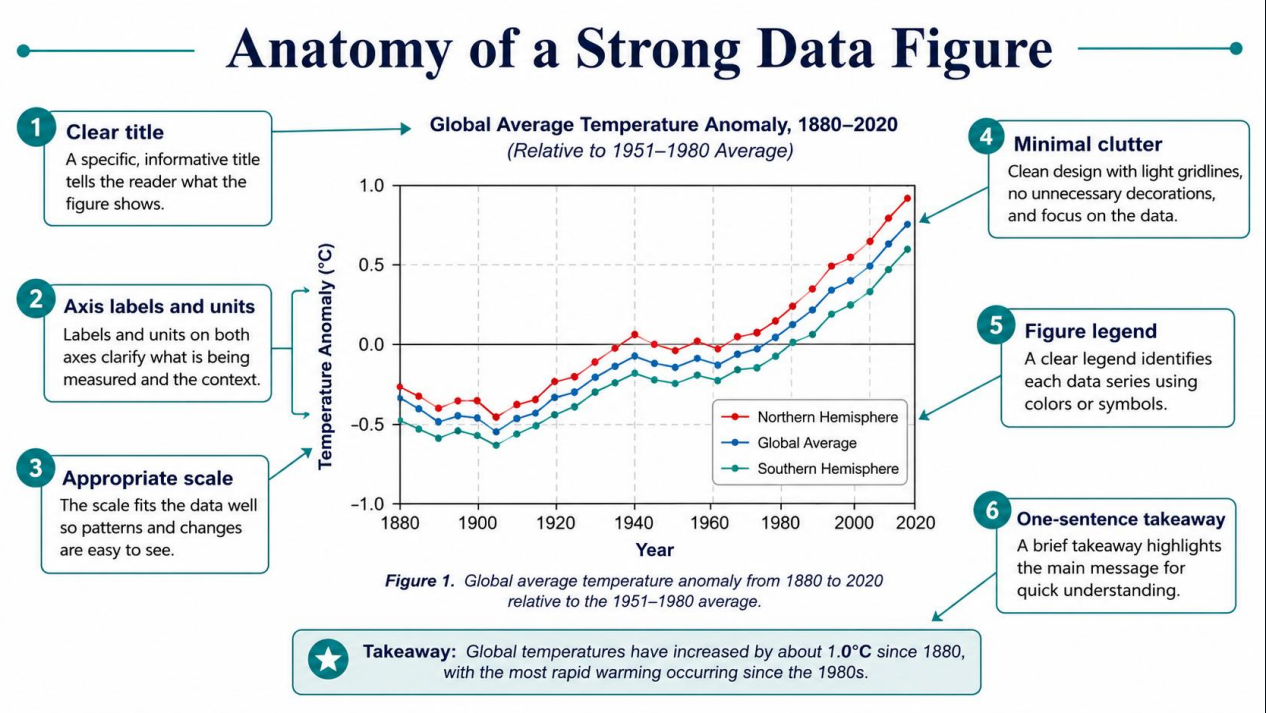


Figure title
State the comparison and the variable.

Caption
Explain what each color, group, and axis means.

Takeaway sentence
One sentence beginning with “This shows that...”

A strong project poster needs 3–4 figures

RECOMMENDED SET

More figures are not automatically better.

Figure 1 Cohort or comparison

- Subtype chart
- or Genomic Alterations comparison

Figure 2 Gene-level evidence

- OncoPrint
- or lollipop plot

Figure 3 Pathway context

- Enrichr top pathways
- or pathway table

Figure 4 Network or outcome

- STRING network
- or survival curve

Each figure should answer a different part of the same question.

How to present cBioPortal results

Translate portal outputs into biological statements.

Do show

- Group names and sample sizes
- Alteration frequency
- Alteration type
- Statistical result when available
- Figure source

Do not claim

- The gene causes the outcome
- The target is validated
- The result applies to all patients
- The figure proves treatment response

**Good result sentence:
“TP53 alterations were
more frequent in
BRCA_Basal than
BRCA_LumA tumors,
suggesting stronger
disruption of tumor-
suppressor pathways in the
Basal-like group.”**

How to present Enrichr and STRING

These tools help explain what a gene list might mean biologically.

Enrichr result should report

- Top pathway or process names
- Adjusted p-value or combined score
- Genes from your list that overlap
- One biological interpretation

STRING result should report

- Whether the proteins form a connected network
- Which nodes look central
- Evidence type is functional association
- One limitation: edges are not always direct binding

Layout controls whether people can follow your poster

DESIGN

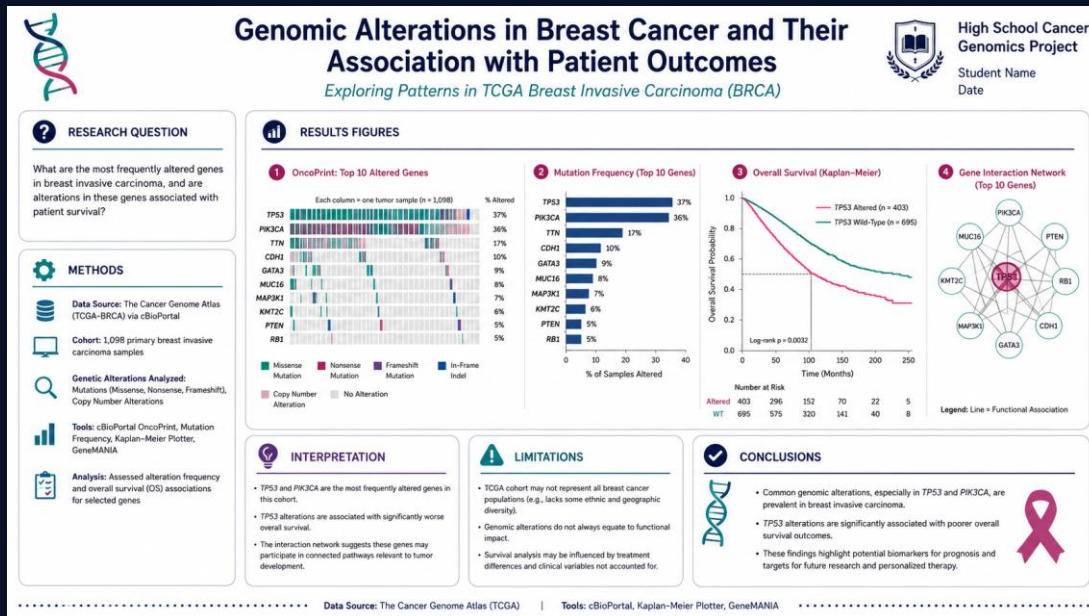
The reader should know where to start, where to look next, and what to remember.

Reading order

- Left to right
- Top to bottom
- Number sections if needed

Visual hierarchy

- Big title
- Large figures
- Short text blocks
- Consistent colors



Poster writing is compressed scientific writing

WRITING

Use fewer words and stronger sentences.

Too wordy

“The cBioPortal database was utilized by our group in order to examine a variety of genes that may be important in breast cancer and there were some genes that seemed to have different alteration frequencies.”

Better

“TP53 alterations were more common in BRCA_Basal tumors than BRCA_LumA tumors, supporting TP53 pathway disruption as a distinguishing feature of Basal-like breast cancer.”

Limitations make the poster stronger

Honest limits show that you understand the science.

Limitations you should consider

- Observational data cannot prove causation
- Subtype groups may have unequal sizes
- Survival may be confounded by stage or treatment
- A database signal is not drug validation

Strong limitation wording

- “This analysis prioritizes a candidate gene; it does not prove therapeutic efficacy.”
- “Survival associations require additional controlled analysis.”
- “The next step would be experimental validation in relevant models.”

Prepare a 2–3 minute poster pitch

PRESENTATION

A good pitch guides the listener through the poster without reading every word.

1. Question

“We asked whether...”

2. Method

“We compared these groups using...”

3. Key result

“The main pattern was...”

4. Meaning

“This suggests...”

5. Limit

“But this does not prove...”

Build your poster framework

Draft

Do this before you open PowerPoint, Canva, or Google Slides.

Your group must write

- One-sentence research question
- Provisional title
- Three figure titles
- One takeaway per figure
- One limitation
- One next step

Editing

- Which figure is your strongest evidence?
- Which figure may need to be removed?
- What claim are you trying to support?
- What claim would be too strong?

Poster checklist before submission

FINAL CHECK

Use this as a final quality-control pass.

Content

- Clear question
- Appropriate dataset and groups
- Figures answer the question
- Limitations included
- Next step stated

Figures

- Readable labels
- Captions explain groups
- No unnecessary screenshots
- Takeaway sentence under each figure
- Consistent colors

Presentation

- Title is specific
- Text is concise
- Reading order is obvious
- Tool and data sources acknowledged
- No causal overclaims